

Madeline K Rowland

formerly Madeline K Jackson

PhD Candidate
Wildlife Biology Program, University of Montana
Population and Disease Ecology Lab
madeline.rowland@umontana.edu

Education

- 2022 – Present **Wildlife Biology PhD Candidate, University of Montana**
Population and Disease Ecology Lab
Advisor: Dr. Angela Luis
Expected Graduation: Spring 2027
Cumulative GPA: 4.0
- 2013 – 2017 **BS, University of Minnesota**
Ecology, Evolution, and Behavior
Advisor: Dr. James Cotner
Cumulative GPA: 3.916

Work and Research Experience

- 2018 – 2022 **Lead Research and Management Assistant - Yellowstone Wolf Project**
Yellowstone Center for Resources
Yellowstone National Park
Mammoth, WY 82190
Supervisors: Drs. Doug Smith and Dan Stahler
- 2018 **Remote Wolf Population Non-invasive Genetic Surveys**
Idaho Department of Fish and Game
Nez Perce – Clearwater National Forests and Selway-Bitterroot Wilderness
Lewiston, ID 83501
Supervisor: Morgan Pfander
- 2018 **Swift Fox Trapping, Telemetry, and Short-Range Collar Data Retrieval**
Jachowski Wildlife Ecology and Restoration Lab – Clemson University
Northeastern Montana
Whitewater, MT 59544
Supervisors: Andrew Butler
- 2017 **Wolf Trapping, Chemical Immobilization, and GPS Collaring**
US Geological Survey Wolf-Deer Study
Boundary Waters Canoe Area Wilderness and Superior National Forest
Ely, MN 55731
Supervisor: Dr. Shannon Barber-Meyer

- 2015 **African Large Mammal Population Distance Sampling**
Carbon Tanzania
Yaeda Valley near Lake Eyasi
School For Field Studies, Rhotia, Tanzania
Supervisor: Dr. Christian Kiffner
- 2015 **Itasca Biological Station Field Research Assistant**
University of Minnesota Itasca Biological Station and Labs
Itasca State Park
Park Rapids, MN 56470
Supervisor: Dr. David Biesboer
- 2013 – 2015 **Limnology Lab Undergraduate Research Assistant**
Cotner Limnology Lab
University of Minnesota
St. Paul, MN 55108
Supervisor: Dr. Jim Cotner

Teaching Experience

- 2026 **Conservation of Wildlife Populations**
Graduate Teaching Assistant
University of Montana
Supervisor: Dr. Mahdieh Tourani
- 2026 **Wildlife Habitat Conservation & Management**
Graduate Teaching Assistant
University of Montana
Supervisor: Dr. Erim Gomez
- 2023 **Conservation of Wildlife Populations**
Graduate Teaching Assistant
University of Montana
Supervisor: Dr. Angela Luis
- 2017 **Animal Diversity Lab – Zoology Instruction via Animal Dissections**
Lead Undergraduate Teaching Assistant and Lab Coordinator
University of Minnesota
Supervisor: Adam Englehardt
- 2015 – 2017 **Wolves and North Woods Carnivores**
Undergraduate Teaching Assistant
Audubon Center of the Northwoods
Supervisor: Bryan Wood

- 2014 – 2017 **Animal Diversity Lab – Zoology Instruction via Animal Dissections**
Undergraduate Teaching Assistant
University of Minnesota
Supervisor: Adam Englehardt
- 2016 **Field Mammalogy at Itasca**
Undergraduate Teaching Assistant
University of Minnesota Itasca Biological Station and Labs
Supervisor: Dr. Joseph Whittaker

Training and Certifications

- 2026 **Modelling Wildlife Aggregation in R**
USGS Workshop, Montana Chapter of the Wildlife Society, Missoula, MT 59812
- 2024 **Structured Decision Making**
Montana Cooperative Wildlife Research Unit, Missoula, MT 59812
- 2024 **Wilderness First Aid**
Aerie, Missoula, MT 59812
- 2023 **Bayesian Integrated Population Modeling using JAGS**
Swiss Ornithological Institute and University of Montana, Missoula, MT 59812
- 2023 **Field Harassment and Assault Prevention Workshop**
FieldFutures, Missoula, MT 59812
- 2022 **Avalanche Safety Training Level 1**
Beartooth Powder Guides, Cooke City, MT 59081
- 2015 **Wildlife Handling and Chemical Immobilization**
Wildlife Science Center/Carlos Avery WMA, Columbus, MN 55025
Instructors: Terry Kreeger and Peggy Callahan

Academic Distinctions and Awards

- 2026 **W.A. Franke Fellowship**
\$20,000 USD, University of Montana Wildlife Biology Program
- 2026 **W.A. Franke Scholarship**
\$5,833.33 USD, University of Montana Wildlife Biology Program
- 2026 **Herrick Graduate Student Travel Award**
\$1,000 USD, University of Montana Wildlife Biology Program
- 2025 **Stella Duncan Memorial Fellowship**
\$28,100 USD, University of Montana Graduate School

- 2025 **Dave Brittell Memorial Scholarship**
\$3,000 USD, Bess Hjartarson and Diane Brittell
- 2025 **Dissertation Improvement Grant**
\$2,500 USD, University of Montana Institute on Ecosystems
- 2025 **W.A. Franke Scholarship**
\$5,512.50 USD, W.A. Franke College of Forestry & Conservation
- 2025 **Herrick Graduate Student Travel Award**
\$1,000 USD, University of Montana Wildlife Biology Program
- 2024 **NSF Graduate Research Fellowship Program**
Honorable Mention
- 2024 **Firefly Grant**
\$2,500 USD, Conservation Nation
- 2024 **John Richard Seiver Scholarship**
\$1,200 USD, W.A. Franke College of Forestry & Conservation
- 2024 **W.A. Franke Scholarship**
\$5,500 USD, W.A. Franke College of Forestry & Conservation
- 2023 **Jack and Kathy Thomas Endowed Graduate Scholarship**
\$2,300 USD, W.A. Franke College of Forestry and Conservation
- 2023 **Best Graduate Student Poster**
\$100 USD, MT Chapter of The Wildlife Society Annual Conference
- 2021 **Conservation Nation 2021 Emerging Conservationist Grant**
\$5,000 USD, Conservation Nation
- 2016 **Monica Tsang and James T. Weatherbee Merit Scholarship in Biology**
\$5,000 USD, University of Minnesota
- 2014 **Ben Thoma Scholarship at Itasca**
\$1,000 USD, University of Minnesota
- 2014 **Girl Scout Gold Award and Scholarship**
\$1,000 USD, Girl Scouts of America
- 2014 **William H. Marshall Field Biology Scholarship**
\$500 USD, University of Minnesota

Publications

* last name changed from Jackson to Rowland in 2024

Published

Bujnakova D, Du Plessis SJ, Falcinelli D, Gundlach C, Holmala K, Keller JS, Khumri M, Korbi J, Mysłajek R, Pieruzek-Nowak S, Paule L, Raundrup K, **Rowland MK**, Sinding M, Tamagnini D, Valtonen M, Aspi J, Kvist L, Hipsley CH (2026) Global drivers of morphological variation in grey wolves. *Diversity and Distributions* 32: e70228.

Brandell EE, **Jackson MK**, Cross PC, Piaggio AJ, Taylor DR, Smith DW, Boufana B, Stahler DR, Hudson PJ (2022) Evaluating noninvasive methods for estimating cestode prevalence in a wild carnivore population. *PLoS ONE* 17(11): e0277420.

Jackson MK, Tatton NR, Smith DW (2022) Wolf Recovery in Yellowstone National Park. In: DellaSala DA, Goldstein MI (Eds.), *Imperiled: The Encyclopedia of Conservation*, vol. 3. Elsevier, pp. 301–312.

Cassidy KA, Smith DW, Stahler DR, Stahler E, Metz M, SunderRaj J, Rabe T, Binder W, **Jackson M**, Packila M, Cassidy B, Rabe J, Tatton N, Meyer C, Bott A, Ho C, Lacey C, Sanborn D (2023) Yellowstone National Park Wolf Project Annual Report 2022. National Park Service, Yellowstone Center for Resources, Yellowstone National Park, WY, USA, YCR-2023-04.

Cassidy KA, Smith DW, Stahler DR, Stahler E, Metz M, SunderRaj J, **Jackson M**, Binder W, Bland T (2022) Yellowstone National Park Wolf Project Annual Report 2021. National Park Service, Yellowstone Center for Resources, Yellowstone National Park, WY, USA, YCR-2022-02.

Cassidy KA, Smith DW, Stahler DR, Stahler E, Metz M, Meyer C, SunderRaj J, **Jackson M**, Binder W, Cassidy B, Rabe J, Tatton N (2021) Yellowstone National Park Wolf Project Annual Report 2020. National Park Service, Yellowstone Center for Resources, Yellowstone National Park, WY, USA, YCR-2021-03.

Smith DW, Stahler DR, Cassidy KA, Stahler E, Metz M, Meyer C, Rabe J, Tatton N, SunderRaj J, Carroll L, **Jackson M**, Cassidy B, Loggers E (2020) Yellowstone National Park Wolf Project Annual Report 2019. National Park Service, Yellowstone Center for Resources, Yellowstone National Park, WY, USA, YCR-2020-01.

Kiffner C, Arndt Z, Foky T, Gaeth M, Gannett A, **Jackson M**, Lellmen G, Love S, Maroldi A, McLaughlin S, Skenandore B, von Euler S, Zambrano Z, Kissui B (2019) Land use, REDD+ and the status of wildlife populations in Yaeda Valley, northern Tanzania. *PLoS ONE* 14 (4): e0214823.

Invited Presentations

Rowland MK. Parasites in the Wild. Parasitology Guest Lecture. University of Montana, 30 October 2025.

Rowland MK and Karnatz C. The Dilution Effect: Lyme Disease Case Study. WILD Health Guest Lecture. University of Montana, 10 February 2025.

Conference Presentations

Rowland MK, Kinsella M, Kuenzi A, Luis AD. Sin Nombre hantavirus interacts with helminth parasite co-infections in Montana deer mice (*Peromyscus maniculatus*). Oral presentation. 74th Annual International Conference of the Wildlife Disease Association, Chicago, IL, 30 July 2026.

Rowland MK, Kinsella M, Kuenzi A, Luis AD. Sin Nombre hantavirus interacts with helminth parasite co-infections in Montana deer mice (*Peromyscus maniculatus*). Oral presentation. 4th Annual Rodent-Borne Zoonotic Disease Workshop, Kilpisjärvi, Finland, 20 June 2026.

Rowland MK, Kinsella M, Kuenzi A, Luis AD. Sin Nombre hantavirus interacts with helminth parasite co-infections in Montana deer mice (*Peromyscus maniculatus*). Poster. Ecology and Evolution of Infectious Disease Conference, Blacksburg, VA, 4 June 2026.

Rowland MK, Kinsella M, Kuenzi A, Luis AD. Sin Nombre hantavirus interacts with helminth parasite co-infections in Montana deer mice (*Peromyscus maniculatus*). Oral presentation. University of Montana GradCon, Missoula, MT, 6 March 2026.

Rowland MK, Kinsella M, Kuenzi A, Luis AD. Sin Nombre hantavirus interacts with helminth parasite co-infections in Montana deer mice (*Peromyscus maniculatus*). Poster. Montana Chapter of the Wildlife Society Conference, Missoula, MT, 26 February 2026.

Rowland MK, Kuenzi A, Luis AD. A First look at parasite co-infection with Sin Nombre hantavirus in Montana deer mice (*Peromyscus maniculatus*). Poster. Ecology and Evolution of Infectious Disease Conference, South Bend, IN, 18 June 2025.

Rowland MK, Kuenzi A, Luis AD. A First look at parasite co-infection with Sin Nombre hantavirus in Montana deer mice (*Peromyscus maniculatus*). Poster. 73rd Annual International Conference of the Wildlife Disease Association, Victoria, British Columbia, Canada, 31 July 2025.

Rowland MK, Keller JS, Smith DW. 3D digitization of Yellowstone wolf skulls for research, education, and outreach. Poster. University of Montana GradCon, Missoula, MT, 7 March 2025.

Jackson MK, Keller JS, Smith DW. 3D digitization of Yellowstone wolf skulls for research, education, and outreach. MT Chapter of The Wildlife Society. Poster. 61st Annual Conference of the Montana Chapter of the Wildlife Society, Helena, MT, 14 February 2023.

Jackson MK, Keller JS, Smith DW. 3D digitization of Yellowstone wolf skulls for research, education, and outreach. Poster. International Wolf Symposium, Minneapolis, MN, 15 October 2022.

Jackson MK, Keller JS, Smith DW. 3D digitization of Yellowstone wolf skulls for research, education, and outreach. Poster. 15th Biennial Scientific Conference on the Greater Yellowstone Ecosystem, Bozeman, MT, 3 September 2022.

Jackson MK, Brandell EE, Smith DW. Environmental factors affecting DNA quality of wolf (*Canis lupus*) fecal samples in Yellowstone National Park. Poster. 100th Annual Meeting of the American Society of Mammalogists, Online, 15 June 2021.

Jackson MK, McNulty KP, Jennelle CS, Fox DL. Patterns of cranial and mandibular shape variation in gray wolves (*Canis lupus*) using geometric morphometric methods. Poster. International Wolf Symposium, Minneapolis, MN, 13 October 2018.

Jackson MK, Iannarilli F, Zoellmer J. Modeling post-reintroduction population dynamics of wolves in Yellowstone National Park. Poster. International Wolf Symposium, Minneapolis, MN, 13 October 2018.

Jackson MK, McNulty KP, Jennelle CS, Fox DL. Patterns of cranial and mandibular shape variation in gray wolves (*Canis lupus*) using geometric morphometric methods. Poster. University of Minnesota Undergraduate Research Symposium, Minneapolis, MN, 22 April 2017.

Jackson MK, Whittaker JC, Thompson SK, Cotner SH. Human influence on antibiotic resistance in the fecal matter of Itasca Rodents. Poster. 96th Annual Meeting of the American Society of Mammalogists, Minneapolis, MN, 28 June 2016.

Public Outreach

2025 **Missoula Public Library spectrUM Event**
Halloween All Under One Roof
October 26, Missoula Public Library

2024 **spectrUM Discovery Area Scientist Poster**
Missoula Public Library
<https://tinyurl.com/mr2p2ntp>

2024 **Missoula Public Library spectrUM Event**
Halloween All Under One Roof
October 27, Missoula Public Library
<https://tinyurl.com/jdt8e6mw>

2024 **Hi-Line Today Radio Interview**
Yellowstone Wolf Skull Digitization
<https://hilinetoday.com/?s=wolf+skulls>

- 2023 **Bozeman Chronicle Interview**
Student passion project to bring Yellowstone wolf skulls online
<http://tinyurl.com/4t63uyh6>
- 2023 **Missoulian Interview**
UM Grad student hopes 3D wolf skulls will allow more people to study the animals she finds 'fascinating'
<http://tinyurl.com/7a53xhyc>
- 2022 **Articulation of Wolf 302M**
Assisted with wolf skeleton articulation for public viewing in Yellowstone
<https://www.yellowstone.org/wolf-302-articulation/>
- 2022 **National Public Radio Interview**
A record number of Yellowstone wolves have been killed. Conservationists are worried.
<https://tinyurl.com/2j2rhzt9>
- 2021 **Yellowstone National Park Instagram: Minute Out In It**
Kleptoparasitism in Yellowstone
<https://tinyurl.com/2p85bh5a>
- 2021 **Wolf Connection Podcast**
Featured Biologist - Episode 30
Minnesota Wolves and Noninvasive Study Methods
<https://tinyurl.com/2dupzr77>